

GOPIKISHAN MAHTO

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PROFESSIONAL SUMMARY

ML Engineer with 3.5 years of experience, including 2+ years building production MLOps pipelines and AI systems. Skilled in object detection, knowledge distillation, LLMs/GenAI, and ONNX/TensorRT deployment. Experienced in optimizing pipelines, automating experimentation, and reducing compute costs.

TECHNICAL SKILLS

AI / ML: PyTorch, Scikit-learn, OpenCV, YOLO, Transformers, Hugging Face, OpenAI, GenAI (Gemini, LLMs), RAG, Knowledge Distillation
MLOps & Infra: MLflow, Wandb, PyTorch DDP (Distributed Training), ONNX, TensorRT, CUDA, Docker, CI/CD, Mixed Precision, NVIDIA Holoscan
Cloud & Data: AWS (EC2, S3), GCP, FAISS (Vector DB), MySQL, Firebase, FastAPI, Django, Pandas, NumPy, Cupy
Languages: Python, SQL, C, C++

EXPERIENCE

Machine Learning Engineer | Endovision AI - New Delhi

Jan 2024 - Present

- Built an agentic ML Research system (Python, Gemini LLMs, YOLO) with researcher, coder, and judge agents that autonomously planned, implemented, and evaluated 150+ training strategies in one week, improving average recall to 70%.
- Distilled a high-capacity EfficientNet-B7 teacher into a lightweight EfficientNet-B5 student via soft-target knowledge distillation, achieving 70% reduction in FLOPs while preserving production accuracy.
- Optimized pipeline latency from 55 ms to 45 ms by minimizing GPU-CPU conversions, applying mixed precision (FP16), and ONNX/TensorRT serialization - achieving 18% latency reduction in production.
- Built an end-to-end automated ML training pipeline - takes user config, runs training, stores artifacts to GCS and logs metrics to Wandb for cross-run comparison and hyperparameter search.
- Built an inference caching pipeline that saved model outputs at multiple stages, enabling post-inference rule tuning without re-running the model - reducing iteration time by 80%.
- Built a device activation tracking system on GCP - Cloud Functions + API Gateway for ingestion, SQL + GCS bucket for logging, data validation layer, and a dashboard for real-time activation visualization.
- Implemented distributed data-parallel training across multiple devices using PyTorch DDP orchestrated by Ray - each device processes different data shards, gradients averaged via AllReduce before weight updates, effectively scaling batch size.

Project Technical Assistant | e-Yantra, IIT Bombay

May 2020 - Jun 2021

- Developed a computer vision pipeline to automatically score competition tasks from recorded video, replacing manual evaluation.
- Controlled a drone using human gestures through computer vision using pose estimation on the ROS framework.
- Published paper 'Real-Time Gesture Control UAV using a low-resource framework' at IRIA IEEE.

KEY PROJECTS

- End-to-End RAG Search Engine: FastAPI + MySQL + FAISS + Transformers with parameterized inference pipeline, user auth, document logs, and LLM results.
- AI Dashboard Generator: Python + Streamlit + LLM pipeline that reads CSVs, generates AI summaries, and auto-creates interactive dashboards with KPIs, filters, and charts.
- Football Goal Detection: YOLO-based model deployed on AWS EC2 to process match footage and auto-save goal clips to S3.

EDUCATION

B.Tech - Computer Science & Engineering - Shri Shankaracharya Technical Campus, Durg

Aug 2021 - May 2024

Diploma - Computer Science & Engineering - UPU Government Polytechnic, Durg

Jul 2017 - Jun 2020

ACHIEVEMENTS

- Ranked 7th out of 1000+ teams in eYRC 2019-20, a robotics competition conducted by IIT Bombay.
- Received Best Paper Award for paper presented at IRIA IEEE hosted by IIT Goa.